Validity Analysis: AFRIDOC_MT

Target context: Ethiopia Ministry of Health — Amharic Clinical Translation Deployment

Overall risk: **HIGH**

Deployment Context

Use case: Translating clinical practice guidelines, vaccination schedules, and maternal-health booklets into Amharic for ministry-of-health distribution

Target population: Health workers and patients across Ethiopian regional health bureaus producing and consuming Amharic-language material

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	3	Moderate gaps	high
Input Content ⚠	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology ⚠	2	Significant gaps	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	4	Minor gaps	high
Average	**2.8**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

AFRIDOC-MT is a thoughtfully constructed ground-up document-level MT benchmark covering Amharic, with strong form match to the Ethiopia MOH deployment (text-only Amharic in Ethiopic script, document-level structure preserved, chrF prioritized for morphological richness, multi-parallel evaluation). However, three of four high-priority dimensions for this deployment have material gaps. Input Content (HIGH priority): the benchmark's WHO-sourced corpus contains substantial geographically remote and Ethiopia-irrelevant content (Afghanistan polio, CCHF, volcanoes, TRIPS), no Ethiopia-specific endemic-disease, calendar, or referral-pathway material, and visible translationese patterns. Output Ontology (HIGH priority): the deployment's primary success criterion — MOH/EFDA glossary compliance — is not operationalized by any benchmark scoring function, and document-level term-consistency is only partially captured. Output Content (HIGH priority): the four named Amharic translators have no documented MOH credentials, clinical backgrounds, or regional affiliations; Krippendorff's α=0.46 for Amharic is the lowest of the five languages; GPT-4o-as-judge is explicitly unreliable for Amharic. The maternal-health and immunization subset of the corpus does provide direct deployment-relevant content [DATASET-D4, D42, D44], and the benchmark's honest documentation of its own limitations (translationese, GPT-4o judge inconsistency, missing document-level QE for Amharic) is itself a significant strength.

ID	Evidence
D4	ተጨማሪ ምግብ ማጠቃሊያ፡- በ 6 ወር አካባቢ የሕፃኑ የኃይል ፍላጎት ... ጨቅላ ህፃናት ... ጡት ማጥባት — Amharic complementary feeding article — maternal-child health topic directly relevant to deployment (maternal-health booklets)
D42	የዓለም ጤና ድርጅት ሕፃናትን ለ6 ወራት ብቻ እንዲያጠቡ ይመክራል ... ደህንነቱ የተጠበቀ እና ተጨማሪ ምግብ — Amharic WHO recommendation on exclusive breastfeeding — direct match for maternal-health booklet content
D44	ጨቅላ ህፃናት ከ6 ወር ጀምሮ የተጣራ፣ የተፈጨ እና ከፊል ጠጣር ምግቦችን መመገብ ይችላሉ — Amharic instruction for pureed/mashed foods from 6 months — clinically precise maternal-health instruction

Practical Guidance

What This Benchmark Measures

AFRIDOC-MT can validly measure (a) general document-level English→Amharic MT fluency and faithfulness on WHO-style health prose using d-chrF/d-BLEU [Q51, Q53], (b) relative model ranking among encoder-decoder NMT and decoder-only LLMs for Amharic [Q4, Q58], and (c) sentence-level vs. pseudo-document tradeoffs revealing that pseudo-document chunking can degrade African-language quality [Q61, Q72]. The benchmark provides a credible signal for whether a system can produce reasonably fluent, faithful Amharic translation of WHO-register health prose at document level.

Construct Depth

Construct depth is shallow for the deployment's primary criteria. Faithfulness and surface fluency are reasonably well covered through d-chrF, human DA, and CE/fluency dimensions [Q183, Q195]. Document-level discourse phenomena are partially probed via lexical and grammatical cohesion error categories [Q189, Q190], though GPT-4o reliability for these is poor in Amharic [Q99, Q219]. The benchmark does NOT probe institutional terminology compliance, clinical-accuracy validation by MOH-credentialed reviewers, regional Amharic register acceptance, or readability for semi-literate patients — all of which are central to deployment success.

What Else You Need

Five concrete supplements are needed: (1) MOH/EFDA glossary-conformance scoring against the Academy of Ethiopian Languages Amharic Medical Dictionary [WEB-8] and any EFDA list — addresses OO gap; (2) parallel reference translations or post-editing by MOH-credentialed clinical translators on a held-out Ethiopia MOH-distributed sample — addresses OC gap; (3) Ethiopia-specific extension corpus covering endemic diseases (malaria, TB, kala-azar, trachoma), Ethiopian calendar dates, and tiered-referral language — addresses IC gap; (4) regional bureau acceptance review with stakeholders from Tigray, Oromia, Somali, Amhara bureaus on register acceptability — addresses OC regional-register gap; (5) readability evaluation with a sample of semi-literate patients or HEW-mediated comprehension testing — addresses OO readability calibration gap.

ID	Evidence
Q4	“Our results indicate that NLLB-200 achieves the best average performance among the standard NMT models, while GPT-4o outperforms general-purpose LLMs.” (p.1)
Q51	“Hence, we realigned sentence-level or pseudo-translation outputs into full documents, then computed BLEU (Papineni et al., 2002) and chrF (Popović, 2015) to create document BLEU (d-BLEU) and document chrF (d-chrF).” (p.5)
Q53	“We report d-chrF scores for the best prompt per model and language direction in the main text, as chrF better captures the morphological richness of African languages (Adelani et al., 2022), with full results provided in Appendix C.” (p.5)
Q58	“On average the NLLB models obtain scores of 65.4/66.6 and 64.3/65.0 on health and tech domains respectively, with 3.3B outperforming 1.3B except when translating into Yorùbá.” (p.5)

Q61	“Pseudo-document translation is worse than sentence-level translation when translating into African languages Our results from pseudo-document translation show a performance drop across different models compared to sentence-level translation, especially when translating into African languages.” (p.6)
Q72	“Human evaluation results align closely with d-chrF, which favors sentence-level translations over pseudo-document translations when translating into African languages.” (p.8)
Q99	“GPT-4o was employed to assess and rate the translation outputs of four models. While its ratings were consistent for translations into English, the same was not observed for translations into African languages, likely due to the model's limited understanding of these languages.” (p.10)
Q183	“GPT-4 is prompted to identify and list the mistakes, such as incorrect translations, omissions, additions, and any other errors, by comparing the model's output to the reference translation.” (p.22)
Q189	“Lexical Cohesion Mistakes: Issues with vocabulary usage, incorrect or missing synonyms, or overuse of certain words that disrupt the flow.” (p.23)
Q190	“Grammatical Cohesion Mistakes: Problems with pronouns, conjunctions, or grammatical structures that link sentences and clauses.” (p.23)
Q195	“Each annotator was assigned 80 documents to evaluate, tasked with marking as many error spans as possible and rating the overall quality on a scale from 0 to 100.” (p.24)
Q219	“These results suggest that using GPT-4o as a translation judge is not yet well-suited for low-resource languages.” (p.26)
WEB-8	Academy of Ethiopian Languages Amharic Medical Dictionary exists as a potentially relevant terminology reference for re-annotation (https://archive.org/details/amharic-dictionary-of-medical-terms-108p-copy)